

Let us consider a Shaft of radius ‘**R**’ with force ‘**F**’ acting tangentially. Let the shaft rotate at **N** rpm due to the turning moment ‘**T = F x d**’. Then, the power transmitted by the circular shaft is calculated by

$$P = \frac{2\pi RN}{60} \text{ Watts}$$

We know that $T = F \times R$,

$$P = \frac{2\pi NT}{60 \times 1000} \text{ kW}$$

Here, T is considered as ‘**Average Torque in Nm**’.

We now have concluded the study on the concepts that are generally related to the design of the shafts. It is necessary to learn the concepts of failure too.

Modulus of Rupture

The concept of a modulus of rupture is synonymous with the concept of a modulus of rupture in bending. According to the definition, this is the "*Ultimate shear stress required to rupture a shaft, calculated using the torsion formula and the empirically determined ultimate torque.*"

$$\text{Mathematically, } \tau_r = \frac{T_u R}{I_p}$$

Here,

τ_r = Modulus of rupture in torsion (also called computed ultimate twisting strength),

T_u = Ultimate torque at failure, and

R = Outer radius of the shaft, m

I_p = Polar moment of Inertia

As the Shear stress is equally distributed at the ultimate torque, rather varying linearly from zero to maximum, it differs significantly from the shearing modulus of rupture.

Despite being hypothetical shear stress, the ‘Shearing Modulus of Rupture’ can be used as a practical tool to determine the maximum torque carrying capacity of shafts using the standard torsion formula.

Let us consider an example problem to understand the concept of the Torsion of Circular Shafts.

Solved Examples on Torsion of Circular Shafts

Q1. A solid circular shaft transmits 75 kW power at 200 rpm. Calculate the shaft diameter, if the twist in the shaft is not to exceed 1° in 2 metres length of the shaft, and shear stress is limited to 50 MN/m^2 . Take $C = 100 \text{ GN/m}^2$

Solution: Given,

$$P = 75 \text{ kW}$$

$$N = 200 \text{ rpm}$$

$$\theta = 1^\circ = 1 \times \frac{\pi}{180} \text{ radian}$$

$$l = 2 \text{ m}$$

$$\tau = 50 \text{ MPa}$$

$$G = 100 \text{ GPa}$$

$$\text{Using Power formula, } P = \frac{2\pi NT}{60 \times 1000}$$

$$\implies 75 = \frac{2\pi \times 200 \times T}{60 \times 1000}$$

$$\Rightarrow T = \frac{75 \times 60 \times 1000}{2\pi \times 200} = 3581 \text{ Nm.}$$

Finding D using Angle of Twist:

Using Torsion Equation, $\frac{T}{J} = \frac{G\theta}{L}$

$$\Rightarrow \frac{3581}{\frac{\pi}{32} \times D^4} = \frac{100 \times 10^9 \times 1 \times \frac{\pi}{180}}{2}$$

$$\Rightarrow D^4 = \frac{3581 \times 2 \times 180 \times 32}{\pi \times \pi \times 100 \times 10^9}$$

$$\therefore D = 0.0804 \text{ m} = 80.4 \text{ mm (Ans.)}$$

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